## Introduction

Lumber cutters need a way to estimate how much lumber a tree contains using measurements that are easy to obtain. In this project, we use the diameter of a ponderosa pine measured at waist height to predict its number of board feet. We compare two models: one assumes that all trees have approximately the same height, while the other assumes that tree height is proportional to diameter. We use the data given to fit each model and compare the accuracy of its predictions.

## Prediction

Before testing the models, we expect the second model to be more accurate because it allows trees with larger diameters to also be taller. The first model assumes that height stays approximately the same regardless of the diameter.

## Variables

$x$ : tree diameter in inches

$y$: number of board feet divided by 10

$h$: tree height

$k$: a constant of proportionality estimated from the data

$V$: tree volume

## Model 1

We approximate each tree as a right circular cylinder and assume all trees are approximately the same height.

Since the diameter is $x$, the radius is $\frac{x}{2}$:

$$V = \pi r^2 h$$

$$= \pi \left(\frac{x}{2}\right)^2 h$$

$$= \frac{\pi h}{4} x^2$$

Because height is approximately constant, $\frac{\pi h}{4}$ is constant. Therefore, **volume is proportional to the diameter squared**. Assuming usable lumber is proportional to volume gives the model

$$y = kx^2$$

To estimate $k$, we use the least-squares formula from Section 3.3 (pg.123). This method selects the constant that minimizes the sum of squared differences between the actual and predicted values. For the power $n = 2$, the formula becomes:

$$k = \frac{\sum {x_i}^2 y_i}{\sum {x_i}^4}$$

Here, $x_i$ and $y_i$ represent the diameter and lumber value for each tree and Σ means adding the values of all ten trees.

Using the given data, our spreadsheet gives:

$$\Sigma {x_i}^2 y_i = 1{,}568{,}429$$

$$\Sigma {x_i}^4 = 9{,}931{,}878$$

Thus,

$$k = \frac{1568429}{9931878} \approx 0.157918674$$

The fitted model is:

$$y = 0.157918674x^2$$

For example, for a tree with a diameter of 17 inches, the predicted value is:

$$y(17) = 0.157918674(17)^2 \approx 45.6385$$

Since $y$ represents board feet divided by 10, this predicts approximately 456.4 board feet. The actual value is 19, or 190 board feet. We then calculate the error as the observed value $y_i$ minus the predicted value $y(x_i)$:

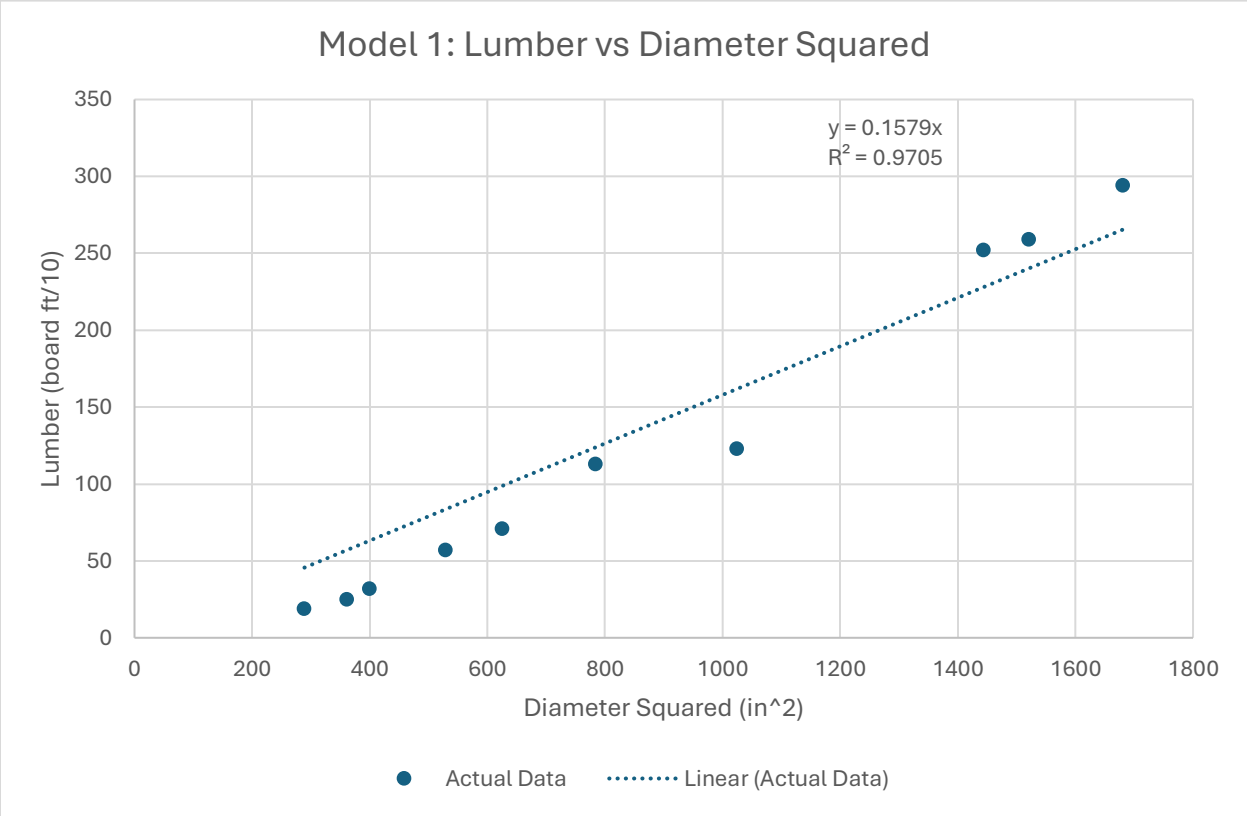
$$y_i - y(x_i) = 19 - 45.6385 \approx -26.6385$$

The negative error means that the model overestimated the amount of lumber. We square each error so that the positive and negative errors do not cancel when added. Squaring also gives larger errors more weight. The least-squares method selects $k$ to make the total squared error as small as possible. For this tree, the squared error is:

$$[y_i - y(x_i)]^2 \approx 709.61$$

We repeat this calculation for all ten trees and add the squared errors. Using the data given gives:

$$\Sigma[y_i - y(x_i)]^2 \approx 7{,}534.77$$




To check the model, we plotted the actual lumber values ($y$) against diameter squared ($x^2$). If the model fits the data well, the points should fall close to the straight line through the origin.

The graph equation, $y = 0.1579x$, uses diameter squared as its horizontal variable, so it matches our model $y = 0.1579x^2$. Excel reports $R^2 = 0.9705$, which suggests a close overall fit when using an intercept of zero.

However, the model tends to overestimate the lumber in smaller trees and underestimate the lumber in larger trees. This suggests that assuming all trees are approximately the same height may not be a good fit for this data set.

**Model 2**

For our second model, we approximate each tree as a right circular cylinder and assume that the height of the tree is proportional to the diameter.

Since the diameter is $x$, the radius is $\frac{x}{2}$. Since height is proportional to diameter, we write $h = cx$, where $c$ is a constant. Substituting this into the cylinder's volume formula gives:

$$V = \pi\left(\frac{x}{2}\right)^2 h$$

$$= \pi\left(\frac{x}{2}\right)^2 (cx)$$

$$= \frac{\pi c}{4} x^3$$

Therefore, volume is proportional to the diameter cubed. Assuming usable lumber is proportional to volume gives us the model:

$$y = kx^3$$

To estimate $k$, we used the least-squared formula for Section 3.3. For the power $n = 3$, the formula becomes:

$$k = \frac{\Sigma {x_i}^3 y_i}{\Sigma {x_i}^6}$$

Using the given data, our Excel spreadsheet gives us:

$$\Sigma {x_i}^3 y_i = 58,288,895$$

$$\Sigma {x_i}^6 = 13,362,776,478$$

Thus,

$$k = \frac{58{,}288{,}895}{13{,}362{,}776{,}478} \approx 0.004362035$$

The fitted model is:

$$y = 0.004362035x^3$$

We can test the fitted model for a tree with a diameter of 17 inches. The predicted value gives:

$$y(17) = 0.004362035(17)^3 \approx 21.4307$$

Since y represents board feet divided by 10, the model predicts approximately 214.3 board feet. The actual value is 19, or 190 board feet. To calculate the error, we do the observed value $y_i$ minus the predicted value $y(x_i)$ :

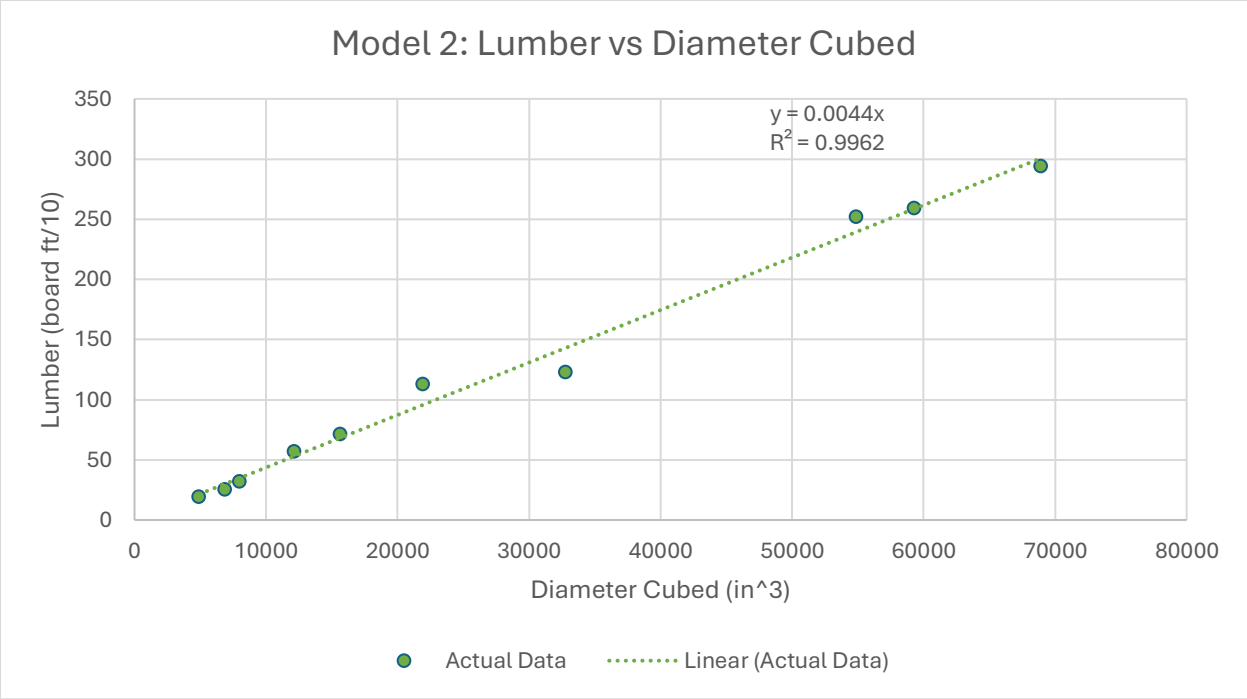
$$y_i - y(x_i) = 19 - 21.4307 \approx -2.4307$$

Since the error is negative, it means that the model overestimated the amount of lumber. We square each error so that the positive and negative errors do not cancel when added. The least-squares method chooses $k$ to make the total squared error as small as possible. For this tree, the square error is:

$$[y_i - y(x_i)]^2 \approx 5.91$$

As in Model 1, we repeat this calculation for all ten trees and add the squared errors. Using the given, Excel gives us:

$$\sum [y_i - y(x_i)]^2 \approx 960.82$$




To check the model, we plotted the actual lumber values ($y$) against diameter cubed ($x^3$). If the model fits the data well, the points on the graph should fall close to the straight line through the origin.

The graph equation, $y = 0.0044x$, uses diameter cubed as its horizontal variable, so it matches our model $y = 0.004362035x^3$. Excel gave us $R^2 = 0.9962$, which is closer to 1 than Model 1's value of 0.9705, suggesting that Model 2 fits the data better.

The points on this graph fall close to the line, suggesting that Model 2, which assumes tree height is proportional to diameter, fits the data well.

## Conclusion

Our results supported our prediction that Model 2 would fit better than Model 1. Model 2 had a total squared error of 960.82, while Model 1 had a much larger error of 7,534.77. This shows that Model 2's predictions were closer to the actual lumber values overall. For this data set, assuming tree height is proportional to diameter gave a better fit than assuming all trees have approximately the same height.

However, these results are based on ten Ponderosa Pine trees with diameters between 17 and 41 inches. The model may work for similar trees, but we would need more data before using it for other species of trees outside this size range.